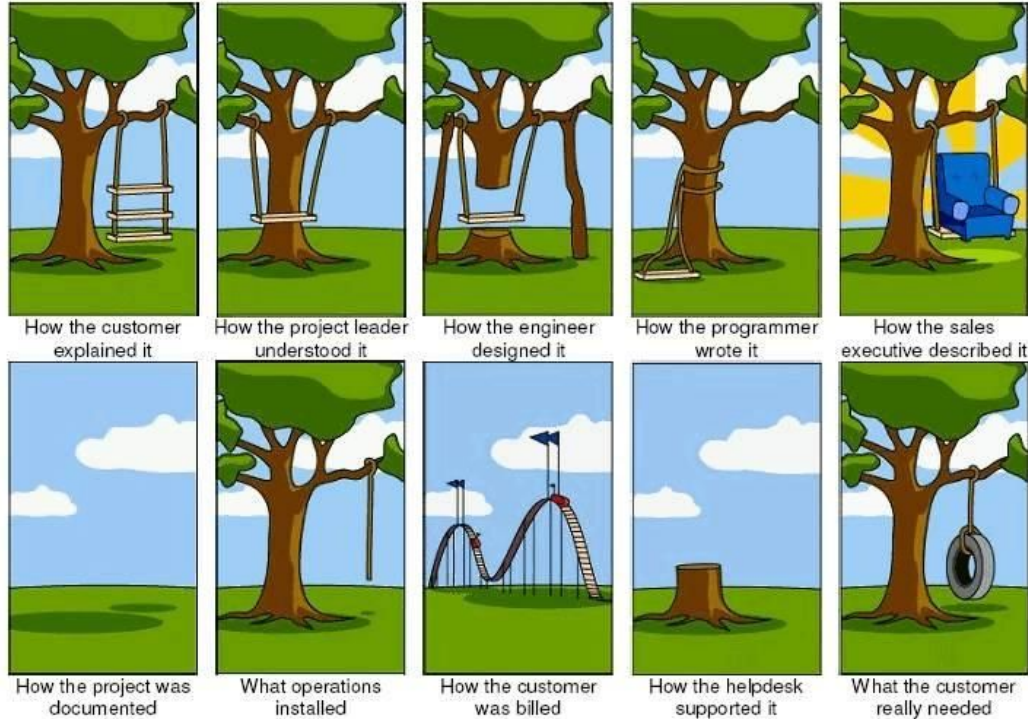


A Large Language Model as a Support Tool for User Story Creation with Use Cases: An Empirical Study

Pedro Garcia, Eduardo Figueiredo, Kattiana Constantino,
Filipe Fernandes and Filipe Roseiro Côgo

Requirements Specification - Problem



The Conextra Template

DESCRIPTION

AS I: <Persona>

I WANT TO: <Goal>

SO THAT: <Reason>

USE CASES (Main or Alternative Flow)

GIVEN THAT: <precondition>

AND ... AND ... <precondition(s)>

WHEN: <actions>

THEN: <postcondition(s)>

DESCRIPTION

AS A: Customer

I WANT TO: Access my account

SO THAT: I can manage my finances

USE CASE (Main Flow)

GIVEN THAT: I have credentials

AND: I entered them correctly

WHEN: I press the button

THEN: Account balance appears

Chat Generative Pre Trained (ChatGPT)

What is it?

- AI language model developed by OpenAI.
- Generate natural language text.
- Trained on large datasets to answer questions, explain concepts, and assist with writing or coding.
- Used in education, research, customer support, and software development.

How to Use ChatGPT

- Go to <https://chat.openai.com>
- Log in with Google, Microsoft, or email.
- Keep default configurations.
- Type a question and press Enter.
- Get instant answers and continue the conversation.

Goal and Research Question

Goal: Investigate the impact of using Large Language Models as a tool in documenting user stories.

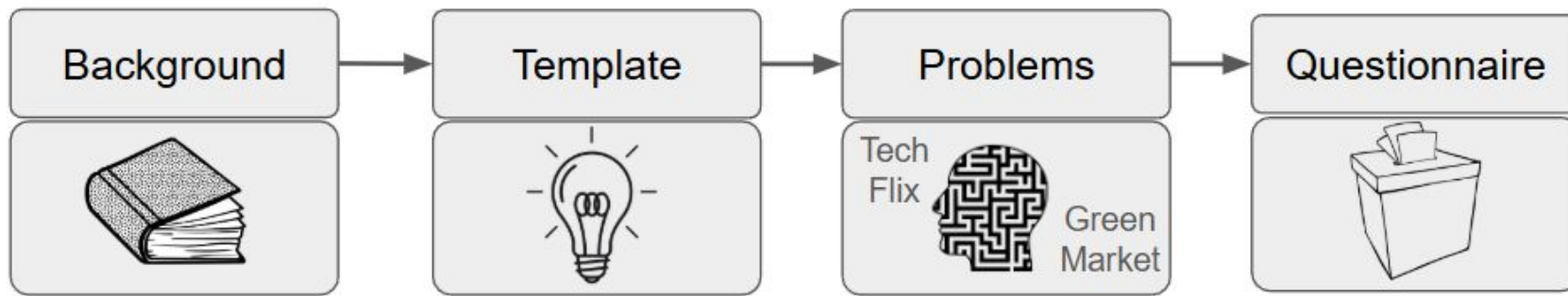
- **RQ1:** Does the use of LLMs impact the time required to create user stories?
- **RQ2:** Does the use of LLMs impact the quality of the documented user stories?
- **RQ3:** What benefits and challenges do developers perceive in using LLMs for requirements documentation?
- **RQ4:** How do developers interact with LLMs when creating user stories and use cases?

Latin Square

1. Division of participants into 2 groups (G1 and G2)
2. Gathering Basic Information
3. Execution

Time	G1	G2
00 - 25	T01 without LLM	T01 with LLM
25 - 50	T02 with LLM	T02 without LLM

Methods



Task Example

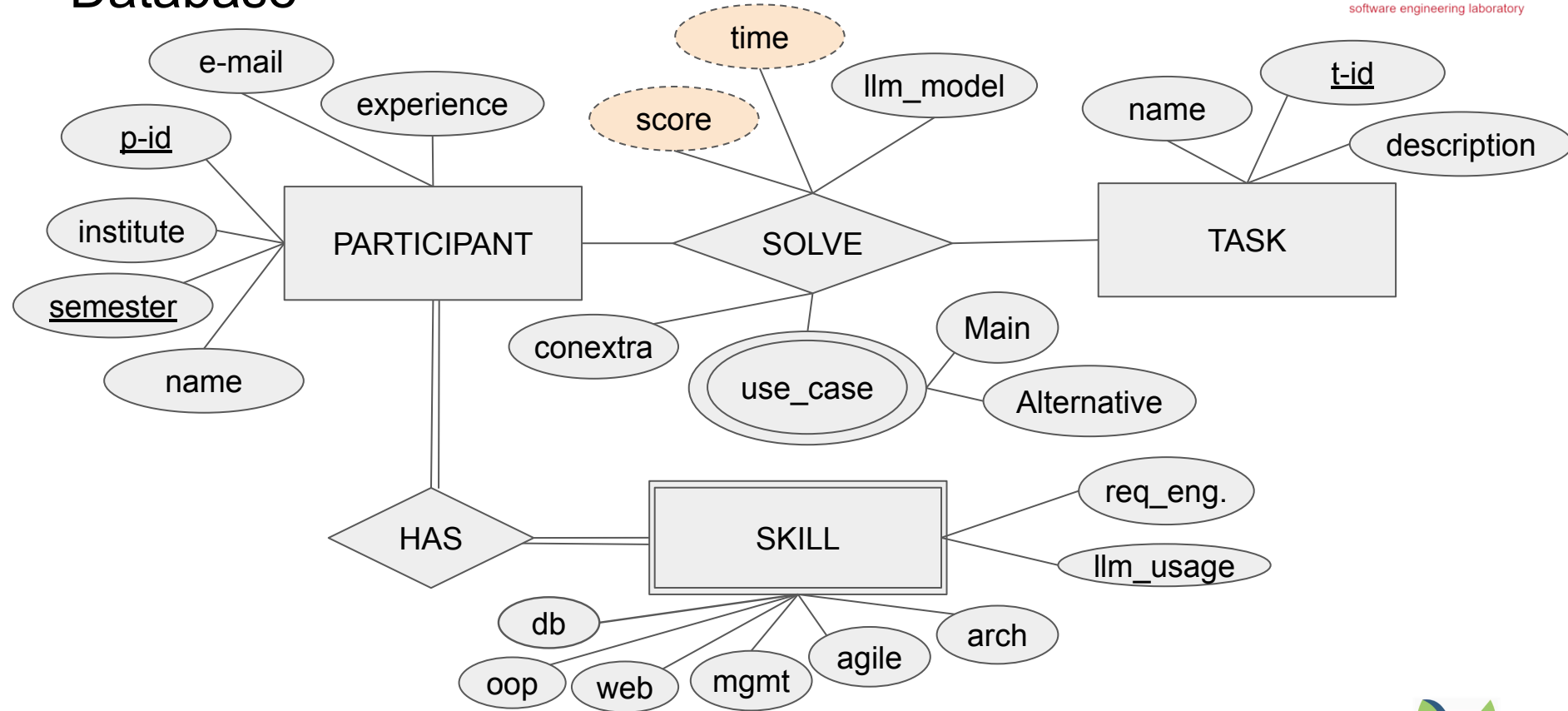
EduFlow

Build a **User Story** with 2 **Use Cases**, including one **Main Flow** and one **Alternative Flow** for the **EduFlow** application described below.

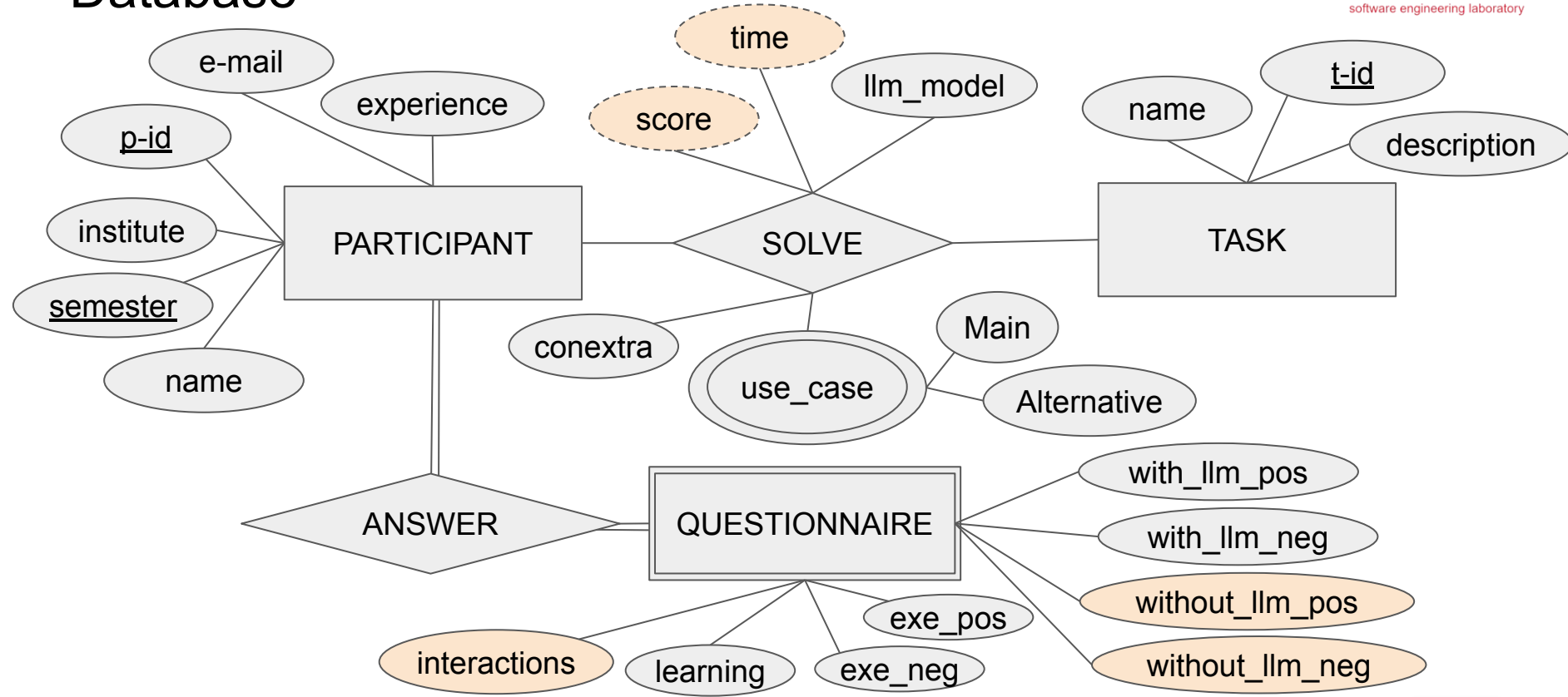
EduFlow is an online platform that recommends personalized learning content and paths for professionals, based on their progress and interests. The system uses artificial intelligence to suggest courses, articles, and practical challenges, encouraging continuous and adaptive learning. Companies can also use the platform to monitor their teams' development, create internal training programs, and assess employees' skill progression.

Possible features: Personalized learning path, gamification, performance evaluation, integration with external courses.

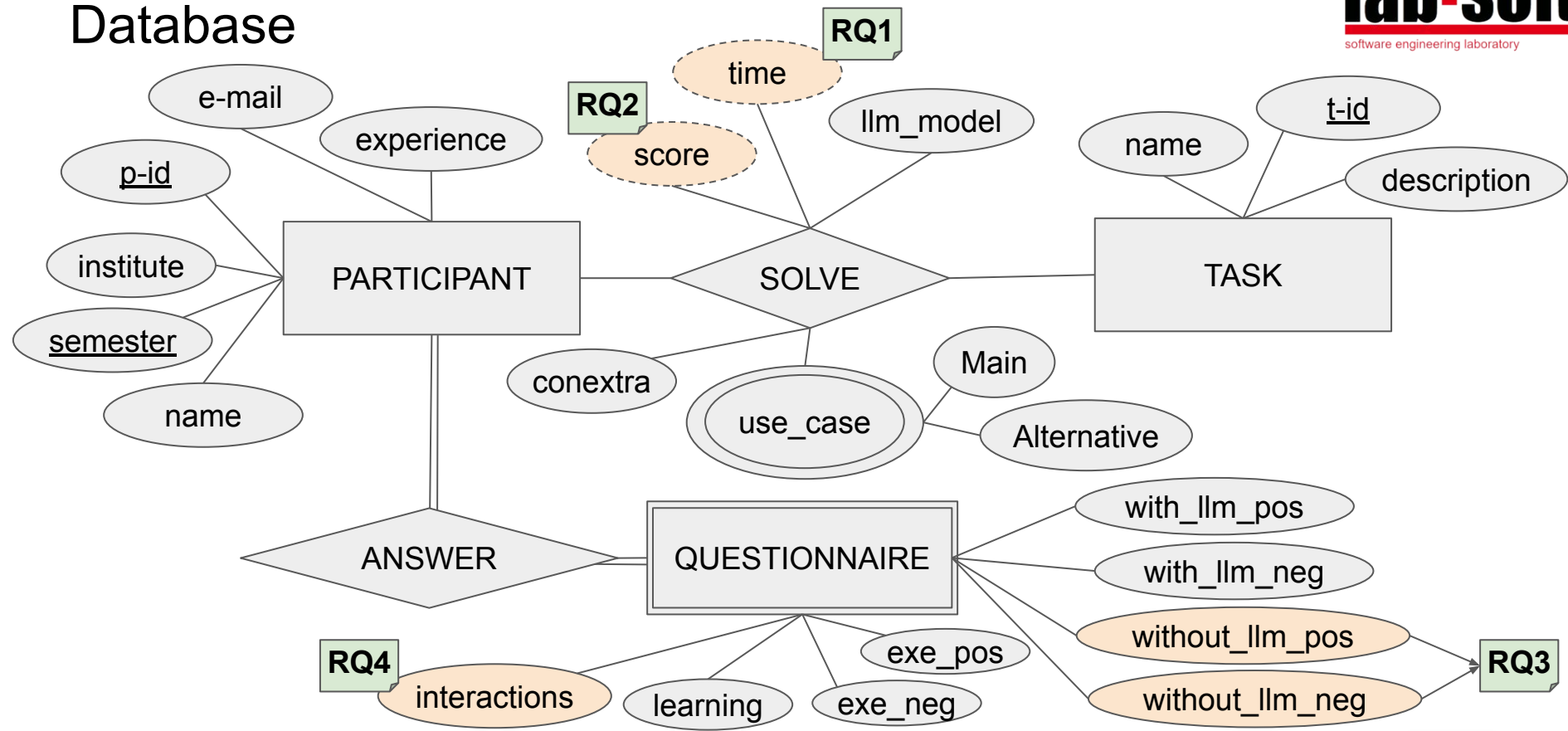
Database



Database

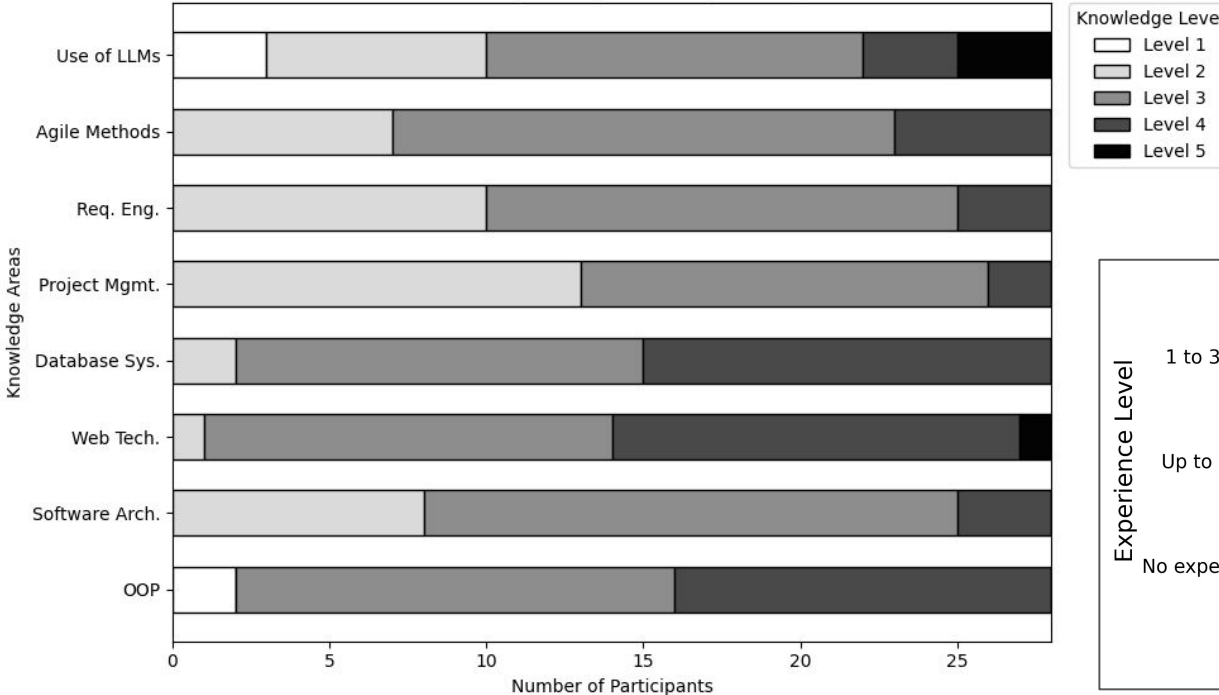


Database



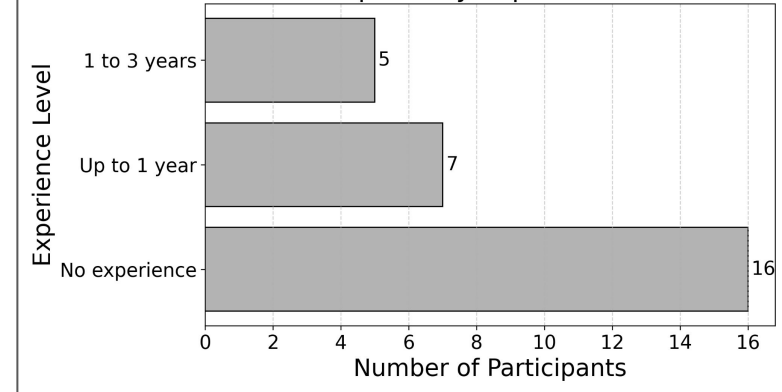
Participants Categorization

Number of Participants by Knowledge Level per Area

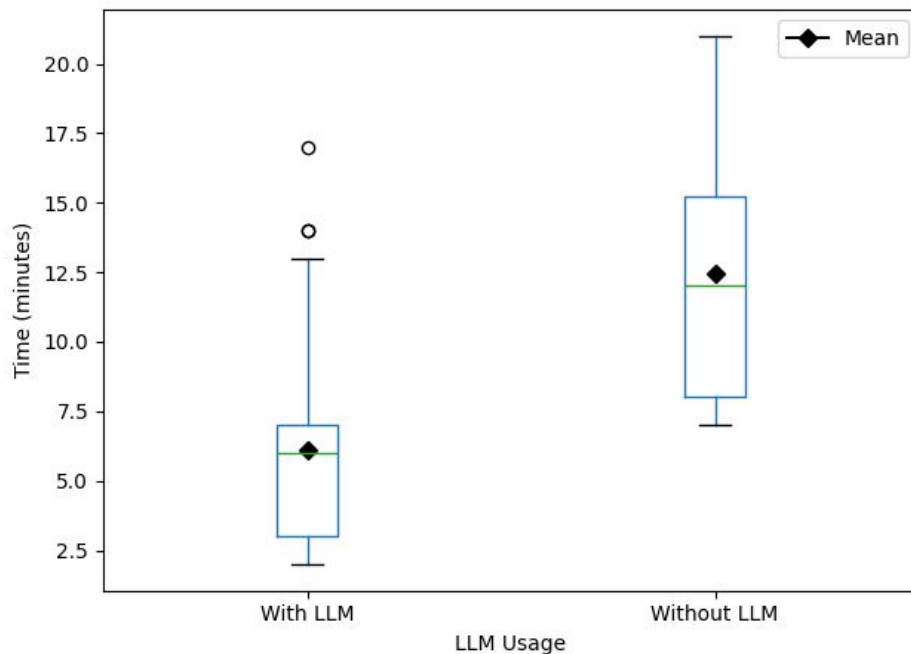


28 participants from two universities.

Participants by Experience Level

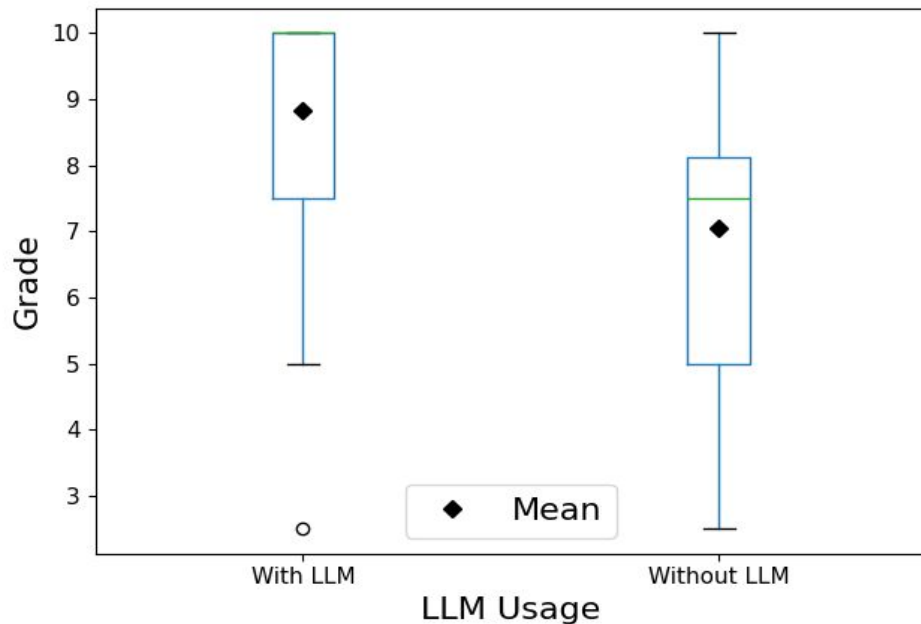


RQ1: Impact on Time



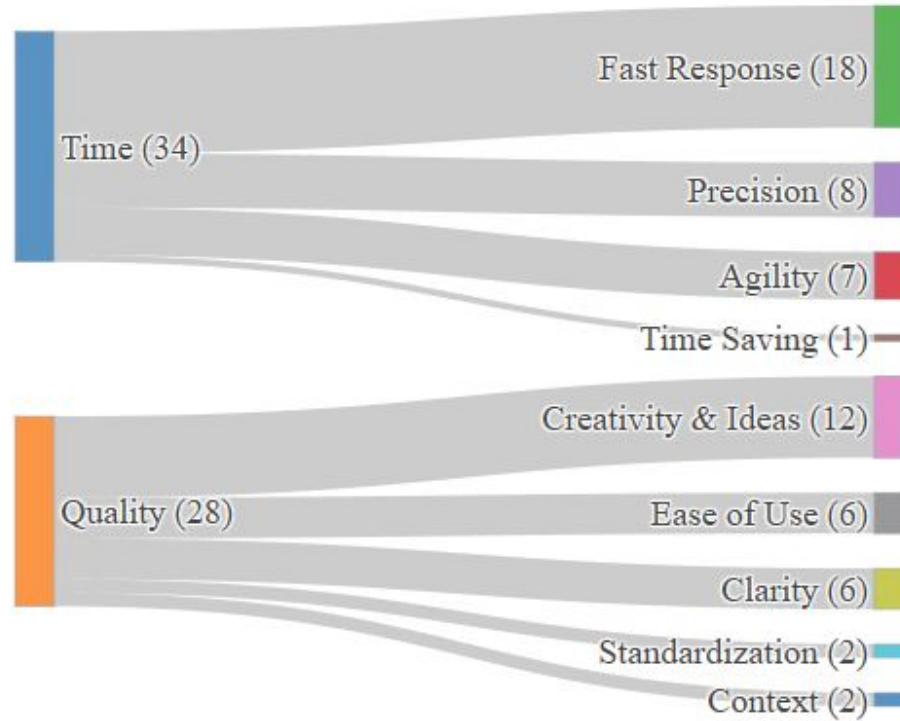
The analysis revealed a reduction in task completion time, **from 12 to 6 minutes** on average, when participants received **support from LLMs**. The Wilcoxon signed-rank test confirmed a statistically significant difference.

RQ2: Impact on Quality

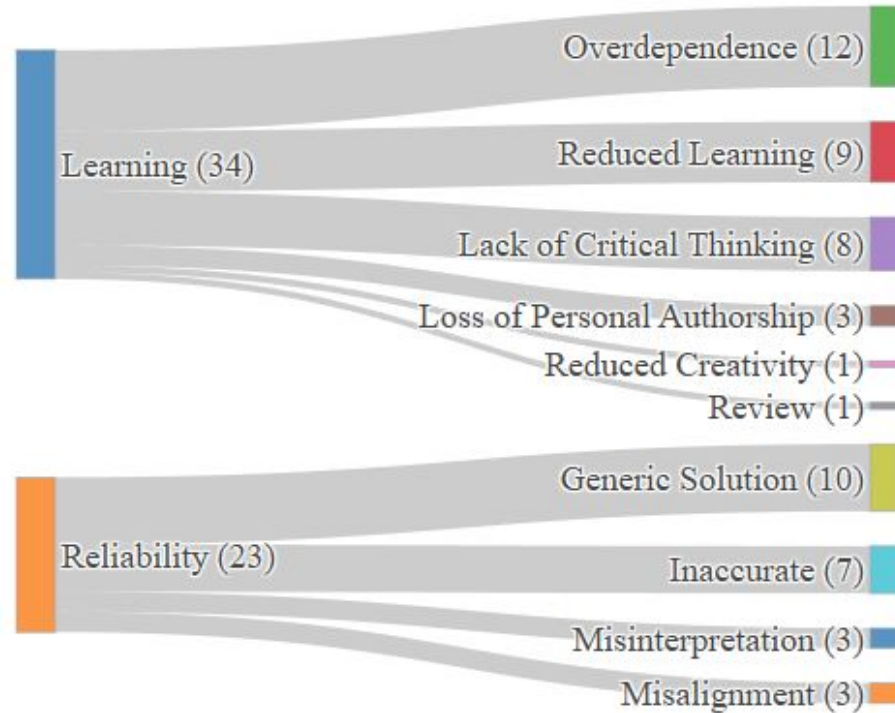


Artifacts produced with **LLM support** **received higher scores** (mean = 8.8) compared to those created without LLM support (mean = 7.0) when evaluated by SE specialists. The Wilcoxon signed-rank test revealed a statistically significant difference in artifact scores between conditions ($p < 0.05$).

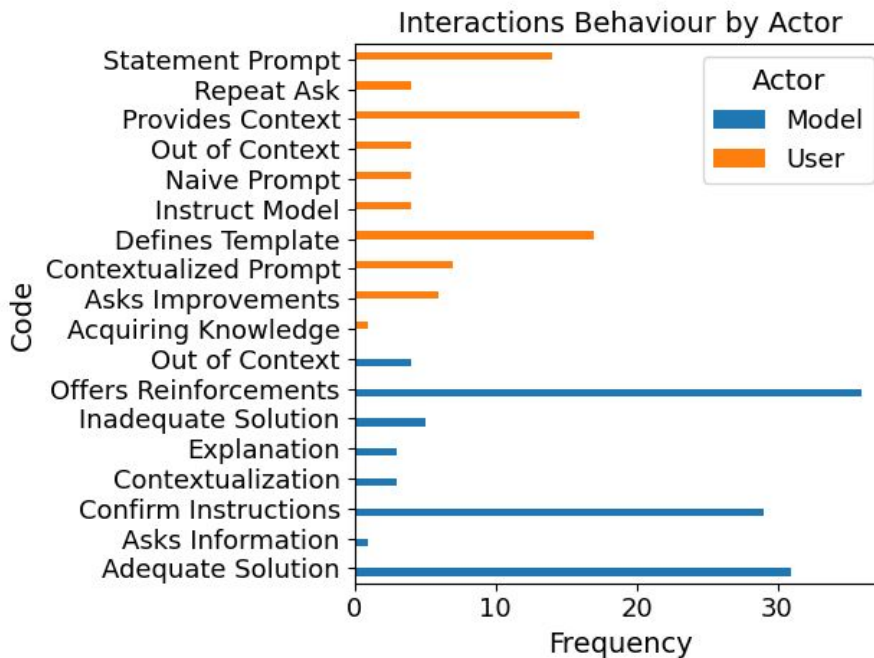
RQ3: Benefits Perceived



RQ3: Challenges Perceived

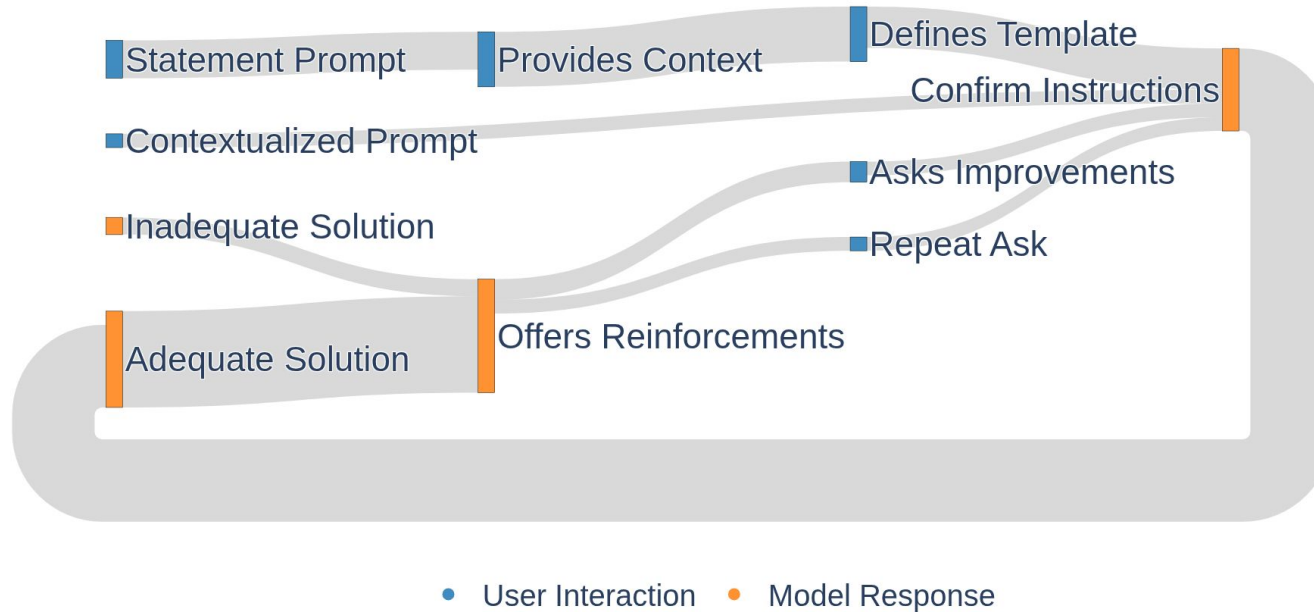


RQ4: Interactions User and LLM

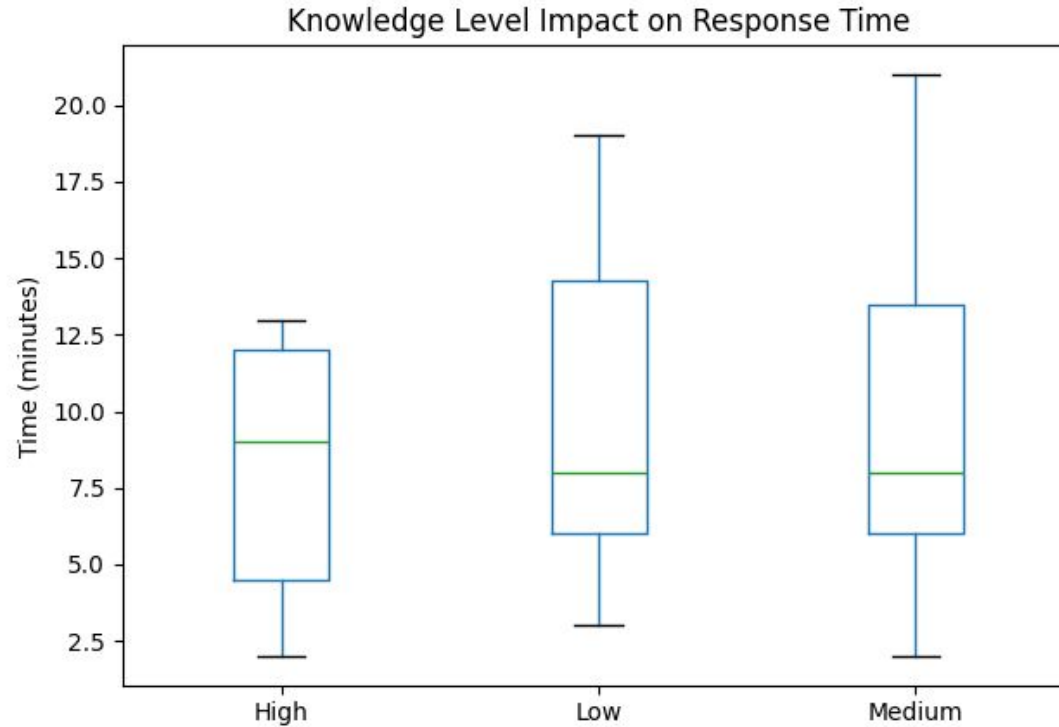


RQ4: Interactions User and LLM

Dominant User-LLM Interaction Flows (RQ4)



Knowledge Level on Requirements



Contributions

Our results show that while LLMs significantly reduce task completion time, this comes at a learning cost.

- (i) LLM-assisted tasks should not replace independent exercises;
- (ii) The quality gain observed must be interpreted cautiously, as higher scores do not necessarily reflect deeper understanding
- (iii) Assessments should evaluate the reasoning process behind requirements, not just the final artifact.

Contributions

Our advice for Professors.

- (i) Introduce LLMs only after students complete tasks
- (ii) Treat prompt engineering as an explicit learning objective
- (iii) Require mandatory critique and revision of LLM outputs
- (iv) Use generic or incorrect LLM-generated user stories as classroom

Contributions

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Contributions

SE Community

- (i) Our controlled Latin Square design combining quantitative and qualitative measures is transferable to other SE education contexts
- (ii) Our RQ4 interaction coding schema provides a reusable framework for studying LLM use in other SE tasks such as code review or testing
- (iii) Our findings on overdependence and reduced critical thinking complement broader SE education concerns raised in the literature

Future Works

- **Increase the number of participants by using other LLMs**
- **Improve the quality analysis by considering criteria beyond the score**
- Conduct a Systematic Literature Review (SLR) and Multivocal Literature Review (MLR) on the use of Language Models (LMs)
- Investigate the use of Small Language Models (SLMs) as an alternative to Large Language Models (LLMs)
- Conduct a survey with domain specialists to validate and contextualize the results
- Evaluate the impact of Language Models on the quality of User Stories (USs)
- Compare learning outcomes between groups with and without Language Model support

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Thanks!

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